

Madhu Pahar, PhD

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Profile

Madhu is an experienced end-to-end AI engineer, data scientist, and researcher in machine learning, signal processing, and multimodal AI for real-world clinical applications. He specialises in translating cutting-edge healthcare research into deployable, production-grade AI systems, with expertise spanning the full AI lifecycle – from data engineering and model development to cloud deployment and interactive user interfaces.

Core Expertise

Signal Processing

Audio · Speech · Sensor · Feature Engineering · Noise-Robust Analysis · ASR (Whisper, Wav2Vec2, HuBERT) · Speaker Diarization · VAD (Silero) · Speech Enhancement · openSMILE · jiwer · Librosa · SoundFile · TorchAudio · PyDub · SoX · FFmpeg · Kaldi · OpenCV · PyFeat

Machine Learning

Deep Learning · Supervised Learning · Classification & Regression · Multimodal AI (Audio + Video + Text) · Transfer Learning · Model Evaluation · Feature Selection · Explainable AI · Time-Series Analysis · Graph Neural Networks · XGBoost · LightGBM · PyTorch · TensorFlow · Keras · Hugging Face · Scikit-learn

NLP & LLMs

Prompt Engineering · Retrieval-Augmented Generation (RAG) · Embeddings & Semantic Search · LangChain · LangFuse · LangSmith · OpenAI · Groq · Jina AI · spaCy · NLTK · Gensim · SentenceTransformers

Healthcare & Clinical AI

Clinical Decision Support · Digital Audio (cough, breath, speech) & Sensor Biomarkers · TB, COVID-19, Cognitive Decline, Mental Health Detection · AI Fairness · Clinical Data

Team Leadership

Project Coordination · Technical Leadership · Data Governance · Stakeholder Management (Clinical, Academic, Industry) · Mentoring & Supervision · Training & SOP Development · Workshop & Event Organisation

Research and Innovation Projects

From Research to Production: A Multimodal AI System for Cognitive Risk Assessment ([Live Demo](#) | [GitHub](#)) May 2025 –

University of Sheffield, UK

- Built and deployed a multimodal AI platform for cognitive risk assessment from clinical speech – combining feature extraction (librosa, spaCy), ML classification (scikit-learn), and LLM-based explainability (Azure OpenAI Phi-4, LangChain).
- Engineered real-time SSE streaming with FastAPI (asyncio, rate limiting) on Google Cloud Run (Docker, CI/CD) and an interactive Next.js frontend (Vercel) with live waveforms and spider-chart visualisation.
- Translated academic research (PROCESS-2/CognoSpeak) into a production-grade, cloud-native application – demonstrating end-to-end AI engineering from data curation to deployment.

Enterprise RAG Platform: Modular AI Assistant System ([GitHub](#)) 2026

Personal Project

- Built a modular RAG platform with multiple domain-specific AI assistants (HR Policy, Inter-speech Research) using LangChain, Groq, FAISS/ChromaDB, and conversational memory – demonstrating modern LLM application engineering.
- Implemented production observability (LangSmith, LangFuse), automated web scraping pipelines (BeautifulSoup), and persistent vector storage with configurable retrieval pipelines for scalable AI assistants.
- Containerised with Docker, designed cloud-ready architecture (Azure/AWS/GCP), and applied software engineering best practices for production-style AI systems.

Automatic detection of cognitive and mental health disorders Jan 2024 –

University of Sheffield, UK

- Develop scalable AI systems to enable early identification of dementia, mild cognitive impairment, depression, anxiety, motor neuron disease, and post-stroke cognitive decline (CognoSpeak, CognoMND and CognoStroke).
- Establish clinically interpretable speech and behavioural biomarkers supporting objective cognitive assessment and clinical decision-making.
- Curate and quality-control longitudinal multimodal datasets from 5,000+ NHS participants in collaboration with clinicians to enable real-world clinical AI research.
- Investigate relationships between attention, memory, and executive function measures and speech-derived behavioural markers of cognitive decline.

COBRE, NIH Project 2023

University of Nevada, Las Vegas (UNLV), USA

- Investigated the relationship of base modifications to disease pathology at the genomic/transcriptomic scale using Oxford Nanopore sequencing and deep learning, and developed a CNN-based solution (NanoCNN).
- Developed computational tools for genomic and transcriptomic data analysis on long-read sequencing data, contributing to biomarker discovery in neuropsychiatric disorders including schizophrenia.

AI for Respiratory Disease Detection & Patient Monitoring 2019 – 2022

Stellenbosch University, South Africa

- Led development of non-invasive AI screening for TB and COVID-19, using cough/breath/speech audio from real clinical environments across South Africa and Uganda, modelling disease severity and hospitalisation risk with large-scale international clinical audio datasets.
- Architected end-to-end clinical data pipelines (audio sensors, mobile platforms, scalable Python workflows) and publicly released rare clinical audio datasets to accelerate global research in AI-based respiratory disease detection at the end of the CAGE-TB project.
- Built patient monitoring solutions using non-invasive smartphone accelerometers for automatic cough detection and bed occupancy monitoring at TASK (24h TB research centre), and developed robust ML models for TB cough classification under noisy real-world conditions.

Core Expertise

Cloud & DevOps

Cloud Architecture (AWS, GCP, Azure)
· AWS (SageMaker, Bedrock, EC2, Lambda, S3) · Azure AI Foundry · Containerisation (Docker) · CI/CD Pipelines (GitHub Actions, Git CLI) · Serverless Computing · Vercel · Render · Linux/Unix

Software & Data Engineering

End-to-End ML Pipelines · API Development (REST, FastAPI) · Asynchronous Programming · Real-Time Streaming (SSE) · Large-scale Data Engineering (ETL)

Frontend & UI

Interactive UI Development · Web Frameworks (Next.js, React) · Tailwind CSS · JavaScript · TypeScript · HTML5 · CSS3

Programming

Python · Anaconda · SQL · Bash · PowerShell · MATLAB · R · Java

Databases

FAISS · ChromaDB · Pinecone · Qdrant · PostgreSQL · SQL · Firebase · Cloud Firestore

Backend/APIs

FastAPI · Django · Flask · REST APIs · Redis · Microservices

Data Science/Engineering

Pandas · NumPy · SciPy · Matplotlib · Seaborn · Ptitprince · Plotly · Openpyxl · Tabulate · n8n

MLOps & Experimentation

MLflow · Weights & Biases (wandb) · TensorBoard · Optuna · Hyperparameter Tuning

GPU & Parallel Computing

CUDA · Multi-GPU Training (PyTorch) · Distributed Data Parallel (DDP) · Mixed Precision Training (AMP) · cuDNN · GPU Optimisation · HPC · Multiprocessing · ThreadPool

Bioinformatics

H5Py · PySAM · Tombo · Guppy · Minimap2 · SAMtools · LongReadSum · ont_fast5_api

AI Coding Assistants

Claude · Codex

Languages

Bengali (mother tongue)
English (fluent)
Hindi (conversational)

Professional Experience

Research Fellow in AI and Speech Processing
University of Sheffield, UK

Jan 2024 –

- Led the development of multimodal ML systems for large-scale clinical conversational speech and video analysis within the CognoSpeak, CognoMND, and CognoStroke programmes, supporting early detection of dementia, motor neuron disease, and stroke-related cognitive impairment in real-world clinical environments.
- Curated, cleaned, and quality-controlled longitudinal multimodal datasets from 5,000+ NHS participants in collaboration with clinicians – integrating cognitive (MoCA) and mental health (PHQ-9, GAD-7) assessments – and publicly released the PROCESS-2 benchmark dataset to accelerate global research in speech-based cognitive impairment detection.
- Designed and deployed production-grade AI systems for real-time clinical decision-support, including speech enhancement, VAD, diarization, ASR fine-tuning, and multimodal feature extraction frameworks (acoustic, prosodic, linguistic, semantic), translating research into clinically actionable tools.
- Developed interpretable biomarker-driven ML models characterising speech and behavioural markers associated with cognitive decline, and investigated algorithmic bias and fairness across multilingual and ethnically diverse UK populations.
- Co-supervised 3 PhD students and disseminated research outcomes through peer-reviewed publications, conference presentations (UKIS 2024, MultiModalAI’24, CMI), and interdisciplinary workshops, bridging the gap between clinical research and deployable AI systems.

Research Scholar in AI and Bioinformatics
University of Nevada, Las Vegas (UNLV), USA

Feb 2023 – Dec 2023

- Developed deep learning frameworks (CNNs, Transformers) for the detection of RNA modifications (5mC, m6A) and variant analysis using Oxford Nanopore long-read sequencing data, contributing to biomarker discovery in neuropsychiatric disorders including schizophrenia.
- Engineered end-to-end bioinformatics pipelines processing raw FAST5 signals into FASTQ/BAM representations, and performed large-scale genomic data analysis (human and yeast) to support model training, benchmarking, and transcriptomic investigation of isoform-level variation and alternative splicing.
- Collaborated with multidisciplinary genomics researchers and supervised graduate/undergraduate students on computational genomics projects, translating complex biological problems into reproducible ML workflows.

Post Doctoral Fellow in AI and Signal Processing
Stellenbosch University, South Africa

Aug 2019 – Jan 2023

- Led development of ML systems for TB and COVID-19 detection from respiratory audio collected in real-world clinical environments across South Africa and Uganda, designing deep learning architectures (attention, transfer learning, generative models) and noise-robust signal-processing workflows.
- Architected and deployed end-to-end clinical data pipelines (audio sensors, mobile platforms, scalable Python workflows) and publicly released the dataset to accelerate global AI research in respiratory disease detection.
- Built patient monitoring solutions using audio sensing and accelerometer data; co-supervised honours/MSc students in pathological speech and cough analysis using reinforcement learning, transformers, and NLP.
- Served as Senior Investigator in the international CAGE-TB study, developed SOPs, trained healthcare workers in AI-enabled sensing technologies, and delivered deep learning training (CNN, LSTM, ResNet) using PyTorch and TensorFlow.

Education

Ph.D. in Computational Neuroscience
University of Stirling, Scotland, UK

2010 – 2016

Thesis: A Novel Sound Reconstruction Technique based on a Spike Code (event) Representation

MSc in Computing for Financial Markets
University of Stirling, Scotland, UK

2009 – 2010

Thesis: Implementation of VaR and Option Strategies

BSc in Mathematics
University of Calcutta, INDIA

2005 – 2009

Mathematics (major), Statistics, Physics

References

Available on request.